# task1

August 7, 2026

[75]:
```
import pandas as pd

train_data = pd.read_csv("SalaryData_Train.csv")
test_data = pd.read_csv("SalaryData_Test.csv")

train_data.dropna(inplace=True)
test_data.dropna(inplace=True)

features = ["age", "workclass", "educationno", "occupation", "capitalgain",␣
 ↪"capitalloss", "hoursperweek"]

X_train = train_data[features]
y_train = train_data["Salary"]

X_test = test_data[features]
y_test = test_data["Salary"]

print("Features:", len(features))
print("Training samples:", len(X_train))
print("Testing samples:", len(X_test))
```

```
Features: 7
Training samples: 30161
Testing samples: 15060
```

[76]:
```
categorical_cols = ["workclass", "occupation"]

X_train_ohe = pd.get_dummies(X_train, drop_first=True)
X_test_ohe = pd.get_dummies(X_test, drop_first=True)
```

[77]:
```
from sklearn.naive_bayes import GaussianNB

model = GaussianNB()
model.fit(X_train_ohe, y_train)
```

[77]:
```
GaussianNB()
```

[78]:
```
y_pred = model.predict(X_test_ohe)
```

[79]:
```
from sklearn.metrics import accuracy_score, classification_report

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 0.796148738379814
Classification Report:
              precision    recall  f1-score   support

       <=50K       0.82      0.94      0.87     11360
        >50K       0.65      0.36      0.47      3700

    accuracy                           0.80     15060
   macro avg       0.74      0.65      0.67     15060
weighted avg       0.78      0.80      0.77     15060
```

[80]:
```
from sklearn.naive_bayes import MultinomialNB

model = MultinomialNB()
model.fit(X_train_ohe, y_train)

y_pred = model.predict(X_test_ohe)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 0.7749667994687915
Classification Report:
              precision    recall  f1-score   support

       <=50K       0.79      0.96      0.87     11360
        >50K       0.62      0.21      0.32      3700

    accuracy                           0.77     15060
   macro avg       0.71      0.58      0.59     15060
weighted avg       0.75      0.77      0.73     15060
```

[81]:
```
from sklearn.naive_bayes import ComplementNB

model = ComplementNB()
model.fit(X_train_ohe, y_train)

y_pred = model.predict(X_test_ohe)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 0.7749667994687915
Classification Report:
              precision    recall  f1-score   support

       <=50K       0.79      0.96      0.87     11360
        >50K       0.62      0.21      0.32      3700

    accuracy                           0.77     15060
   macro avg       0.71      0.58      0.59     15060
weighted avg       0.75      0.77      0.73     15060
```

[82]:
```
from sklearn.naive_bayes import BernoulliNB

model = BernoulliNB()
model.fit(X_train_ohe, y_train)

y_pred = model.predict(X_test_ohe)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 0.7699867197875166
Classification Report:
              precision    recall  f1-score   support

       <=50K       0.81      0.90      0.86     11360
        >50K       0.55      0.37      0.44      3700

    accuracy                           0.77     15060
   macro avg       0.68      0.63      0.65     15060
weighted avg       0.75      0.77      0.75     15060
```

[83]:
```
from sklearn.naive_bayes import CategoricalNB

model = CategoricalNB()
model.fit(X_train_ohe, y_train)

y_pred = model.predict(X_test_ohe)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 0.8350597609561753
Classification Report:
              precision    recall  f1-score   support

       <=50K       0.84      0.96      0.90     11360
```

| | | | | |
|---|---|---|---|---|
| >50K | 0.78 | 0.45 | 0.57 | 3700 |
| | | | | |
| accuracy | | | 0.84 | 15060 |
| macro avg | 0.81 | 0.71 | 0.74 | 15060 |
| weighted avg | 0.83 | 0.84 | 0.82 | 15060 |